

SITE-SPECIFIC MANAGEMENT

Different Techniques to Identify Management Zones Impact Nitrogen and Phosphorus Sampling Variability

Jiyul Chang, David E. Clay,* Charles G. Carlson, Sharon A. Clay, Douglas D. Malo, Robert Berg, Jon Kleinjan, and William Wiebold

ABSTRACT

The efficiency of the management zone approach to improve fertilizer recommendations relies on accurately locating zone boundaries. The objective of this study was to determine the impact of different techniques of identifying management zones on soil NO_3^- -N and Olsen-P (sodium bicarbonate extractable-P) sampling variability. Soil samples were collected on a 60 by 60 m or denser grid, in three fields (65, 53, and 40 ha). These samples were analyzed for NO_3^- -N and Olsen-P. Soil nutrient data was used to simulate the effect of different techniques to identify P and N management zone boundaries. Approaches evaluated for locating management zone sampling boundaries included: (i) sampling areas impacted by old homesteads or animals separately from the rest of the field; (ii) sampling different grid cells; (iii) use of geographic information systems (GIS) or cluster analysis to identify zones based on apparent electrical conductivity (EC_a), elevation, aspect, and distance (connectedness); and (iv) sampling each soil series separately. An F statistic was used to determine if the sampling approach reduced nutrient sampling variability. Results suggested that: (i) old homesteads or areas impacted by animals should be sampled separately from the rest of the field; (ii) grid-cell sampling was more consistent in reducing within zone soil-test variability than the other techniques tested; and (iii) zones that are not continuous should be sampled and managed separately.

A KEY PRECISION AGRICULTURE CONCEPT is that fertilizer recommendations can be improved by accounting for in-field nutrient variability (Chang et al., 2000). Commonly used strategies for obtaining soil nutrient spatial information are grid-point, grid-cell, and management-zone sampling. In grid-point sampling, samples are collected from specific points within fields, and kriging or some other estimation approach is used to develop contour maps (Wollenhaupt et al., 1994; Chang et al., 1999). Grid sampling provides excellent soil nutrient information for land managers if the points are close enough to assure spatial dependence. However, developing a universal recommendation for grid distance is difficult because different fields may have different sampling requirements. For example, Wollenhaupt et al. (1994) had one recommendation for fields in the nonre-

sponsive categories (91-m grid) and a different recommendation for fields in the responsive range (61-m grid). Other problems with grid-point sampling are that important information may be missed when grid distances are too large, and many farmers perceive that grid-point sampling is not profitable.

Grid-cell sampling is an approach where a composite sample is collected from a block with a specified size (Wollenhaupt et al., 1994). The sample from each block is analyzed and the resulting value represents the average value of the cell. Many current traditional soil sampling strategies contain some aspect of grid-cell sampling. Ferguson et al. (1998) and Buchholz (1999) recommended that in Nebraska and Missouri, respectively, the largest sampling area in a field should be 8 ha or less. In Montana, Jacobsen (1998) recommended that the largest sampling area should be 40 ha.

The management zone approach is based on the hypothesis that a field is a mosaic of different habitats with each having unique characteristics that influence soil properties and management (Doerge, 1999; Fleming et al., 2000). Information that can be used to identify the different management zones include (i) prior experience, (ii) yield maps, (iii) soil survey maps, (iv) topography, (v) soil drainage, (vi) EC_a , (vii) soil color, (viii) soil organic matter, (ix) soil nutrients, (x) remote sensing, and (xi) soil moisture. One of the first techniques used to identify different zones was to spread as many maps as possible on a table and to draw lines around the different zones. For example, Fleming et al. (1999) used aerial photographs as templates for farmer-developed productivity management zones in two center-pivot irrigated fields located in Colorado. In these fields, nutrient concentrations (NO_3^- , K, and Zn) and yields were high in farmer-identified high productivity areas, and soil organic matter, percentage clay, and EC_a were correlated to yield.

More recently, researchers have investigated if computer classification procedures available in GIS or cluster analysis can be used to improve and automate the process of defining management zones. For example, Fridgen (2000) and Fridgen et al. (2000) used cluster analysis to identify zone boundaries. Fridgen et al. (2000) reported that in Missouri, approximately 54% of the yield variation was explained by cluster analysis of EC_a , elevation, and slope information. Once boundaries

J. Chang, D.E. Clay, C.G. Carlson, S.A. Clay, D.D. Malo, and J. Kleinjan, Plant Science Dep., South Dakota State Univ., Brookings, SD 57007; R. Berg, Southeast Research Farm, South Dakota Agric. Exp. Stn., South Dakota State Univ., Beresford, SD 57004; W. Wiebold, Univ. of Missouri, Columbia, MO 65211. South Dakota Exp. Stn. paper no. 3337. Received 26 Sept. 2002. *Corresponding author (david_clay@sdstate.edu).

Published in *Agron. J.* 95:1550–1559 (2003).
© American Society of Agronomy
677 S. Segoe Rd., Madison, WI 53711 USA

Abbreviations: EC_a , apparent electrical conductivity; GIS, geographic information systems; MZA, Management Zone Analyst software; NRCS, National Resource Conservation Service; s_p^2 , pooled variance.

are identified, samples from each zone can be collected, analyzed, and used for developing management recommendations. Questions that need to be resolved with attributed-based soil sampling include (i) should samples from two areas with similar attributes, that are not physically connected, be composited?; (ii) should large management zones be split into several smaller zones?; and (iii) what classification procedures should be used to identify management zones?

Several studies have compared different approaches to define zone boundaries. Mallarino and Wittry (2001) evaluated the impact of management zone sampling (digitized soil survey and targeted soil sampling) on fertilizer recommendations. They reported that targeted soil sampling correctly fertilized a higher percentage of the fields than schemes based on field averages and digitized soil survey maps. Franzen et al. (1998) compared topography-based sampling with grid sampling. They reported that NO_3^- -N and P concentrations from management zone demarcation based on topographic information were correlated to values calculated on a 60-m grid. Franzen et al. (2002) compared published soil survey, Order 1 soil survey (scale 1:6600), topography-based, and grid-sampling approaches and reported that published soil surveys should not be used for identifying N management zones unless the soil patterns were verified with other zone development tools. Additional studies are needed to compare traditional approaches for collecting soil samples with computer classification of soil attributes. The objective of this study was to determine the impact of different techniques for identifying management zones on soil NO_3^- -N and Olsen-P sampling variability.

MATERIALS AND METHODS

Study Fields and Soil Samples

This research was conducted at Moody (65 ha), Brookings (53 ha), and Beresford (40 ha) sites. All sites were located in eastern South Dakota and had a crop sequence of corn (*Zea mays* L.) followed by soybean [*Glycine max* (L.) Merr.]. The Moody field was located at 44°10' N, 96°37' W. The Brookings field was located at 44°14' N, 96°39' W. The Beresford site was located at 43°03' N, 96°53' W. Dominant soils at Moody were Cubden (fine-silty, frigid Aeric Calciquoll), Waubay (fine-silty, mixed, superactive, frigid Pachic Hapludoll), Kranzburg (fine-silty, mixed, superactive, frigid Calcic Hapludoll), and Vienna (fine-loamy, mixed, superactive, frigid Calcic Hapludoll). Dominant soils at Brookings were Barnes (fine-loamy, mixed, superactive, frigid Calcic Hapludoll), Brookings (fine-silty, mixed, superactive, frigid Aquic Hapludolls), McIntosh (fine-silty, mixed, superactive, frigid Aeric Calciquoll), and Vienna. Dominant soils at Beresford were Egan (fine-silty, mixed, superactive, mesic Udic Haplustolls) and Chancellor (fine, smectitic, mesic Vertic Argiaquoll). Additional details about the soils at these sites are available in Clay et al. (2001).

Soil samples from the 0- to 15- and 15- to 60-cm depths were collected at Moody from a 30- by 30-m slightly offset grid before planting corn in 1995. At Brookings, soil samples from the same depths were collected from a 60- by 30-m slightly offset grid before planting soybean in 1997. At Beresford, soil samples from the same depths were collected from a 60- by 60-m slightly offset grid following soybean harvest

in October 1997. Each grid sample consisted of 15 individual cores that were collected within 1 m of the center of the grid (Clay et al., 1997). Each sampling point was located by walking a prescribed number of steps from the previous sampling point. Relative to the edge of the field, the starting points for each transect were slightly different. The net result of this sampling approach was grid points that were slightly offset. The sampled points were located by latitude, longitude, and elevation by carrier phase differential global positioning system (DGPS) equipment. The vertical and horizontal error of the system was ≤ 2 cm (Johansen et al., 2001).

Soil samples were prepared for analysis by air-drying (35°C) and grinding to <2 mm in diameter. Olsen-P was extracted from samples collected from the 0- to 15-cm soil depth with 0.5 M NaHCO_3 at a pH value of 8.5 (Olsen and Sommers, 1982). The soil extract was filtered, a color reagent containing ascorbic acid and molybdate was added, and color development was measured on a colorimeter set at 882 nm (Murphy and Riley, 1962). Inorganic N (NH_4 -N and NO_3 -N) was extracted from samples collected at the 0- to 15- and 15- to 60-cm soil depths with 1.0 M KCl with a 10:1 solution to soil ratio and analyzed on an Astoria Analyzer 300 (Astoria-Pacific Inc., Clackamas, OR) by the Cd reduction method (Maynard and Kalra, 1993).

Apparent electrical conductivity in the vertical configuration was measured with an EM38 (Geonics Ltd., Mississauga, ON, Canada) at each sampling point by placing the instrument in the vertical configuration on the ground and recording the measured value (Clay et al., 2001). At all sites, EC_a was collected in the spring of 1997 following thawing and before planting and fertilizing. Black-and-white aerial photographs of the sites (1950–1965) were obtained from county USDA National Resource Conservation Service (NRCS) offices. The IKONOS satellite collected false color composite images (blue, red, and near infrared) at Brookings and Moody on 17 May 2001. This image has a spatial resolution of 4 m.

Locating Boundary Lines between Management Zones

Techniques for locating boundary lines were based on (i) sampling areas impacted by old homesteads or animals separately from the rest of the field; (ii) sampling different grid cells; (iii) use of GIS (Environmental Systems Research Institute, 1996) or cluster analysis (USDA-ARS, 2000) to identify zones based on EC_a , elevation, aspect, and distance (connectivity); and (iv) sampling each soil series separately. All simulated sampling approaches used the same database and the simulated P and N concentrations were equal to the average value of grid samples contained within a zone.

Locating Old Homestead or Animal-Impacted Areas

Old aerial photographs (1950–1985) along with other evidence were used to locate old homesteads and areas where animals had been grazed. The other evidence included the existence of old windmills, trees, haystacks, or entry roads into the fields. On the basis of this classification, each field was split into two areas: old homestead or areas impacted by animals, and the remaining portion of the field. Only one area impacted by homesteads or animals was identified in each field. Aerial photographs taken of the Brookings field in 1956 showed the location of an old homestead (Fig. 1), and an aerial photograph of the Moody field collected in 1984 revealed the existence of haystacks. This area of the field was also adjacent to a homestead and therefore most likely had been grazed.

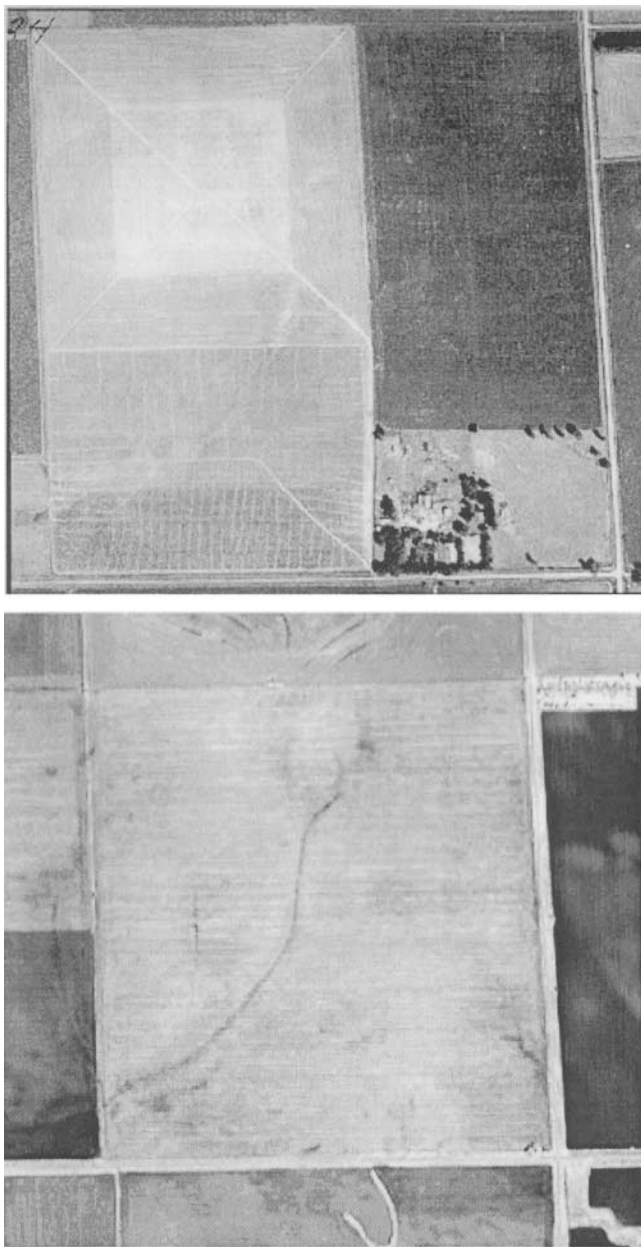


Fig. 1. Black-and-white images (top) collected from the Brookings, SD, field in 1956 and (bottom) a near-infrared image collected in May 2001.

Grid-Cell Sampling

At Moody, the field was split into 16 (4 ha), 9 (7 ha), and 4 (16 ha) square grid cells. At Brookings, the field was split into 14 (3.5 ha) and 8 (6.6 ha) square cells. At Beresford, the field was split into 10 (4 ha), 6 (7 ha), and 3 (13 ha) square cells. The N and P concentration of each grid cell was assigned the average value of all the grid soil samples collected within the block.

Geographic Information Systems Classification

ArcView GIS (Environmental Systems Research Institute, 1996) was used to characterize EC_a , elevation, aspect, and distance information into management zones. For the EC_a -elevation classification, 10 EC_a -elevation zones were identified in a two-step process. In the first step, the fields were

split into five different elevation zones. Each zone had a range that was approximately equal to the difference between the maximum and minimum divided by five. These five zones were then separated into areas that were greater than or less than the average EC_a value. For classification based on EC_a and aspect, the fields were separated into five different equal EC_a classes by the same approach as described above. The EC_a classes were subdivided by aspects (316° to 45° , 46° to 135° , 136° to 225° , and 226° to 315°).

For classification based on EC_a and distance (nonconnectivity), a two-step process was to identify management zones. In the first step, the field was separated into five different equal EC_a classes. Each zone had a range that was approximately equal to difference between the maximum and minimum EC_a value divided by five. In the second step, each large group was further subdivided into areas that were connected or not connected. If the groups were connected, they were identified as a single zone. If the groups were not connected, they were characterized as two zones.

Cluster Analysis Classification

Mahalanobis distance and fuzzy *c*-means unsupervised clustering algorithms within Management Zone Analyst 1.0 (MZA) were used to identify different clusters (Johnson, 1998; Fridgen, 2000). The software is available at USDA-ARS (2000). The combination of three variables (EC_a , elevation, and aspect) was used by MZA. The number of management zones selected for characterization was associated with the minimum Normalized Classification Entropy and Fuzziness Performance Index values (Fridgen, 2000). The number of zones selected was dependent on the data used in the classification. For comparative purposes, the management zone of selected approaches at Moody are shown in Fig. 2. Similar maps were developed for the other fields (data not shown).

Soil Series

Order 1 soil survey maps (scale < 1:12 000) were prepared by the USDA-NRCS staff at Moody and Brookings (Soil Survey Staff, 1993). An Order 1 soil survey was not available at Beresford, and therefore, soil characterization at Beresford was not conducted.

Statistics and Calculations

Summary statistics, skewness values (Ott, 1977), and semi-variograms for Olsen-P and NO_3^- -N were determined. The models used to describe the relationship between sampling distance and the semivariances were exponential, spherical, and linear. The model that accounted for the most variation was the criterion for selecting a given model.

Once management zone boundaries were identified, the mean and variance (s_i^2) of each zone was calculated. These variances were used to calculate pooled variances (s_p^2) by the equation

$$s_p^2 = \sum_{i=1}^z (n_i - 1) s_i^2 / [\sum_{i=1}^z (n_i) - z] \quad [1]$$

where z was the number of sampling zones, n_i was the number of samples within zone i , and s_i^2 was the variance within zone i (Steel and Torrie, 1980). An F value (s_{field}^2/s_p^2) at $P \leq 0.1$ was used to determine significant differences.

The P recommendation for corn from each zone was calculated by the equation

$$\text{P recommendation (kg ha}^{-1} \text{ P)} = (0.7 - 0.044 \times \text{STP}) \times \text{YG} \times 7.86 \quad [2]$$

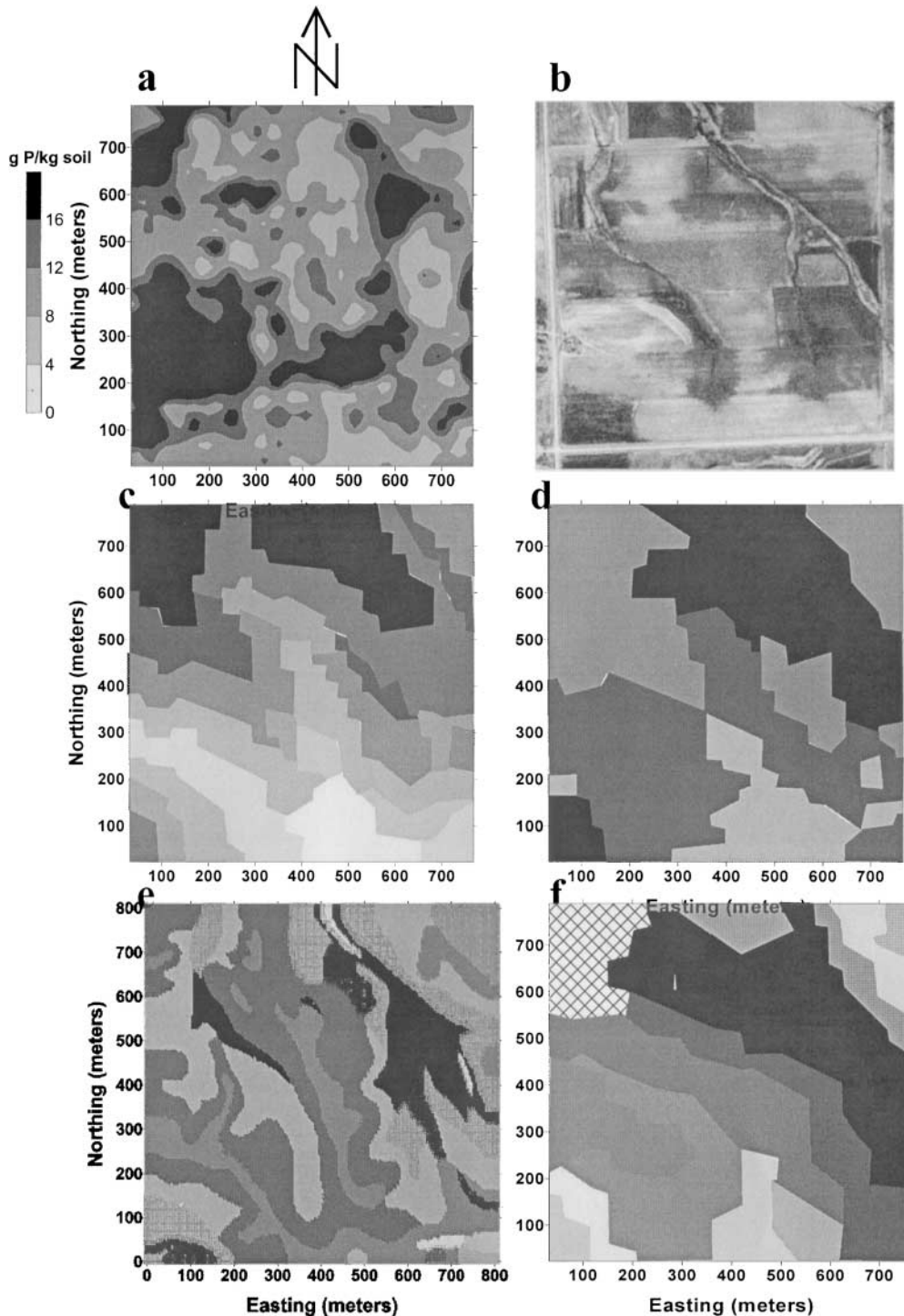


Fig. 2. The selected data layers and classification maps for Moody, SD, are: (a) Olsen-P contour map, (b) black-and-white aerial photo collected in 1984, (c) management zones based on geographic information systems (GIS) analysis of electrical conductivity (EC) and elevation information, (d) management zones based on cluster analysis of EC and elevation information, (e) management zones based on an Order 1 soil survey, and (f) management zones based on GIS analysis of EC and distance information. For c, d, e, and f each color represents a different management zone.

where STP is the soil test P (mg kg^{-1} P) and YG is the yield goal (10.67 Mg ha^{-1}) at 15.5% moisture (Gerwing and Gelderman, 1998). The N fertilizer recommendation for corn was calculated by the equation

$$\text{N recommendation (kg ha}^{-1} \text{ N)} = 21.42 \times \text{YG} - \text{STN} - \text{PCC} \quad [3]$$

where STN is $\text{kg NO}_3\text{-N}$ contained in the surface 61 cm of soil and PCC is the previous crop credit (e.g., legume credit, $44.8 \text{ kg ha}^{-1} \text{ N}$) (Gerwing and Gelderman, 1998).

At each grid point within a management zone, the fertilizer recommendation was compared with the grid point fertilizer recommendation. For P, if the difference between the management zone P recommendation and the grid P recommendation

was < -5 or > 5 kg ha⁻¹ P, then the grid point was characterized as over- or underfertilized, respectively. If the difference was ≥ -5 and ≤ 5 kg P ha⁻¹, then the site was characterized as correctly fertilized. For N, if the difference between the management zone N recommendation and the grid N recommendation was < -10 or > 10 kg ha⁻¹ N, then the grid point was characterized as over- and underfertilized, respectively. If the difference was ≥ -10 and ≤ 10 kg ha⁻¹ N, then the site was characterized as correctly fertilized.

RESULTS AND DISCUSSION

If the precision and accuracy of soil tests are influenced by nutrient variability, then it makes logical sense that sampling techniques with low variability will have higher fertilizer recommendation reproducibility than sampling techniques with high variability. To evaluate the relationships among sampling approach, soil nutrient variability, and fertilizer recommendations, the paper was divided into three sections: (i) Olsen P and nitrate-N population distributions and spatial variability (field characteristics); (ii) the effects of management zone sampling on Olsen-P and resulting fertilizer recommendations; and (iii) the effects of management zone sampling on nitrate-N concentrations and resulting fertilizer recommendations.

Field Characteristics

Olsen-P and NO₃⁻-N concentrations had skewness values > 0 and medians less than the means (Table 1). These results indicated that both Olsen-P and NO₃⁻-N were not normally distributed. Kravchenko and Bullock (1999) had similar results for fields located in Illinois, Indiana, and Iowa. Soil Olsen-P concentrations were greater in old homestead or grazing areas than the remaining portions of the field (Table 1). Franzen et al. (1998) had similar results in a field located near Valley City, ND, where P level patterns were consistent be-

tween years, and very high P levels were observed where livestock were fed until 1960.

In all fields, nugget-to-sill ratios for Olsen-P and NO₃⁻-N were relatively small, which indicated strong to moderate spatial dependence (Cambardella et al., 1994). Spatial dependence is important and suggests that as the distance between the sampling points increases, the NO₃⁻-N and Olsen-P values became less correlated.

Soil Olsen-Phosphorus

At Moody, removing the sampling points within the old homestead or grazing area from the whole field data set reduced the variance (Table 2). Splitting the field further into 4-ha grid cells or classifying the site by GIS with EC_a-distance information reduced s_p^2 . Classification based on only EC_a information (data not shown) or EC_a-elevation or EC_a-aspect did not reduce s_p^2 . These results suggest that at this site, nutrient spatial dependence is important and that areas not physically connected should not be composited. Use of cluster analysis or the Order 1 soil survey to characterize management zones did not reduce s_p^2 .

At Brookings, sampling the old homestead separately from the whole field reduced s_p^2 63% (Table 3). The s_p^2 was further reduced 26% by splitting the field into 3.5-ha grid cells. Cluster, GIS, or sampling by soil series did not reduce s_p^2 .

At Beresford, sampling the old homestead or areas impacted by animals separately from the whole field reduced s_p^2 95% (Table 4). Pooled variance values were further reduced by subdividing the field into 4- and 7-ha grid cells. Classification based on soil attribute information had mixed results. When the data set contained areas impacted by either an old homestead or animals, s_p^2 was reduced by GIS (EC_a-elevation, EC_a-aspect, and EC_a-distance) and cluster analysis (EC_a-aspect, EC_a-elevation-aspect) approaches. When areas impacted by

Table 1. The means, minimums, maximums, medians, variances, skewnesses of the whole field, and whole field without the old homestead or grazing areas. The exponential (exp), spherical (sph), and linear (lin) models were used to develop the semivariograms.

	Moody 1995		Brookings 1997		Beresford 1997	
	Olsen P	NO ₃ ⁻	Olsen P	NO ₃ ⁻	Olsen P	NO ₃ ⁻
	mg kg ⁻¹					
Whole field						
Mean	13.3	12.3	18.1	4.4	25.4	10.2
Median	11.0	9.2	15.0	3.9	12.0	8.0
Variance	60.8	72.6	268	8.1	1611	54
Skewness	1.7	1.1	6.2	3.6	4.2	4.2
No. of samples	536		352		100	
Field w/o homestead						
Mean	12.3	12.6	16.4	4.3	13.9	9.4
Median	11.0	9.4	15.0	3.8	11.0	7.7
Variance	39.2	74.6	97.6	6.3	86.9	18.1
Skewness	1.4	1.0	4.5	2.8	1.9	1.2
Homestead						
Mean	33.1	6.9	61.3	7.4	114	16.6
Variance	62.1	4.6	2775	42.5	4778	301
Area, ha	4		2.3		5.3	
	Semivariogram					
Nugget	27.1	23.5	21.4	3.9	25.1	8.0
Sill	74.9	101	248	9.1	120	20.0
Range, m	795	784	1572	824	420	320
Nugget/sill	0.4	0.2	0.1	0.4	0.2	0.4
Model	exp	sph	sph	lin	sph	sph

Table 2. The influence of sampling approach on Olsen-P (mg kg^{-1}) pooled variances (s_p^2) at Moody, SD. The information used in classification approaches were combinations of apparent electrical conductivity (EC_a), elevation (elev.), distance, and aspect.

Sampling methods	Classes (<i>n</i>)	No. of zones	Old homestead or animal-impacted area			
			Sampled separately		Not sampled separately	
			s_p^2	<i>F</i> value	s_p^2	<i>F</i> value
Grid cell						
4-ha grid cell	16	16	32.7	1.20†	42.5	1.43†
7-ha grid cell	9	9	36.7	1.07	50.5	1.20†
16-ha grid cell	4	4	37.1	1.06	51.8	1.17
GIS‡						
EC_a -elev.	10	29	36.2	1.08	53.5	1.14
EC_a -aspect	18	62	35.3	1.11	50.4	1.20†
EC_a -distance	15	15	33.9	1.16†	45.1	1.35†
Cluster						
EC_a -elev.	4	10	38.8	1.01	57.5	1.06
EC_a -aspect	2	19	39.2	1.00	57.1	1.06
EC_a -elev.-aspect	5	8	39.2	1.00	60.9	1.00
Order 1 soil survey	16	40	38.2	1.03	59.4	1.02
Whole field	1		39.2		60.8	

† s_p^2 values are significantly different from the whole field variance at $P = 0.1$. For the numerator, $\text{df} = 536$. For the denominator, $\text{df} = 536 - n$.

‡ GIS, geographic information systems.

either a homestead or animals were sampled separately, management zones based on GIS and cluster analysis did not influence s_p^2 .

At Moody, Brookings, and Beresford, sampling areas impacted by a homestead or animals separately from the rest of the field reduced the field variance 36, 63, and 95%, respectively. These results show that simply sampling old homesteads separately from the rest of the field should reduce Olsen-P sampling variability, which in turn should improve P recommendations. These findings are in agreement with many current soil sampling recommendations (Jacobsen, 1998; Buchholz, 1999; Kleinjan, 2002).

In the three fields, when the homestead or grazing area was sampled separately, splitting the field into 4-ha (or 3.5-ha) grid cells further reduced s_p^2 . These results were attributed to spatial dependence. Strong spatial dependence (nugget-to-sill ratio < 0.25) was observed in Brookings and Beresford, while in Moody only moderate (nugget-to-sill ratio between 0.25 and 0.75) spatial dependence was observed (Table 1). Strong and moderate spatial dependence indicates that as the sampling points become farther away, Olsen-P values become less correlated. The importance of distance on reducing

nutrient variability was supported by the observation that the only classification approach that further reduced s_p^2 (two of three sites) was GIS classification of EC_a -distance information. These results suggest that zones that are not continuous should be sampled and managed separately.

Including areas impacted by a homestead or animals in the data set reduced the whole field P recommendation and increased the amount of the field that was underfertilized at Moody (Table 5). Splitting the field into 4-ha grid cells increased fertilizer rates when compared with the whole field recommendation. When the homestead or animal-impacted areas were sampled separately, splitting the field into 4-ha grid cells reduced overfertilized areas from 31.5 to 27.6% and reduced underfertilized areas from 57.1 to 45.0%. Similar improvements were observed when the old homestead was not sampled separately.

At Brookings, the whole field P recommendation was 0 kg P ha^{-1} . Grid-cell sampling increased P recommendations (Table 5), increased the overfertilized portion of the field and reduced the underfertilized portion of the field. At Beresford, sampling areas impacted by either an old homestead or animals separately from the

Table 3. The influence of sampling approach on Olsen-P (mg kg^{-1}) pooled variances (s_p^2) at Brookings, SD. The information used in classification approaches were combinations of apparent electrical conductivity (EC_a), elevation (elev.), distance, and aspect.

Sampling methods	Classes (<i>n</i>)	No. of zones	Old homestead or animal-impacted areas			
			Sampled separately		Not sampled separately	
			s_p^2	<i>F</i> value	s_p^2	<i>F</i> value
Grid cell						
3.5-ha grid cell	14	14	71.7	1.36†	189	1.42†
6.6-ha grid cell	8	8	81.0	1.21†	211	1.27†
GIS						
EC_a -elev.	9	24	87.7	1.11	251	1.07
EC_a -aspect	13	56	94.2	1.04	251	1.07
EC_a -distance	10	10	91.0	1.07	248	1.08
Cluster						
EC_a -elev.	8	34	91.3	1.07	251	1.07
EC_a -aspect	9	67	90.3	1.08	244	1.10
EC_a -elev.-aspect	8	38	90.0	1.08	249	1.08
Order 1 soil survey	8	37	92.3	1.06	264	1.02
Whole field	1		97.6		268	

† s_p^2 values are significantly different from the whole field variance at $P = 0.1$. For the numerator, $\text{df} = 352$. For the denominator, $\text{df} = 352 - n$.

Table 4. The influence of sampling approach on Olsen-P (mg kg^{-1}) pooled variances (s_p^2) at Beresford, SD. The information used in classification approaches were combinations of apparent electrical conductivity (EC_a), elevation (elev.), distance, and aspect.

Sampling methods	Classes (n)	No. of zones	Old homestead or animal-impacted areas			
			Sampled separately		Not sampled separately	
			s_p^2	F value	s_p^2	F value
Grid cell						
4-ha grid cell	10	10	57.6	1.51†	1326	1.21
7-ha grid cell	6	6	63.1	1.38†	1292	1.25
13-ha grid cell	3	3	83.5	1.04	1561	1.03
GIS						
EC_a -elev.	8	14	77.0	1.13	1256	1.28†
EC_a -aspect	11	35	81.9	1.06	1108	1.45†
EC_a -distance	11	11	69.9	1.24†	1241	1.30†
Cluster						
EC_a -elev.	8	19	78.1	1.11	1452	1.11
EC_a -aspect	5	20	71.3	1.22	930	1.73†
EC_a -elev.-aspect	9	22	77.4	1.12	926	1.74†
Whole field	1		86.9		1611	

† s_p^2 values are significantly different from the whole field variance at $P = 0.1$. For the numerator, $\text{df} = 100$. For the denominator, $\text{df} = 100 - n$.

whole field increased the whole field recommendation. Grid-cell sampling further increased the P recommendation and reduced the percentage of land both under- and overfertilized.

Nitrate-Nitrogen

At Moody, sampling areas impacted by either an old homestead or animals separately from the rest of the field did not reduce the NO_3^- -N field variance (Table 6). However, s_p^2 was reduced by grid-cell sampling (4, 7, and 16 ha) and GIS classification of EC_a -distance information. Cluster analysis or classification based EC_a -elevation or EC_a -aspect did not reduce s_p^2 . At Brookings, sampling the old homestead separately from the whole field reduced the NO_3^- -N field variance from 8.1 to 6.3 (Table 7). The s_p^2 was further reduced by 3.5-ha grid-cell sampling. Classification based on soil attributes did not reduce NO_3^- -N s_p^2 . At Beresford, sampling the homestead separately from the rest of the field reduced the variance from 54 to 18 (Table 8). After removing the homestead from the database, the only approach that reduced s_p^2 further was 4-ha grid-cell sampling. When the homestead or animal-impacted areas were not sampled separately from the rest of the field, the only approach that reduced s_p^2 was GIS classification based on EC_a -aspect information.

The net effect of grid-cell sampling at Moody resulted

in an increased N recommendation which was in contrast with the two other sites (Table 9). In spite of this increase, the amount of N fertilizer applied to overfertilized areas was greatest (582 kg N per field) for the whole field and least (371 kg N per field) for the 4-ha grid-cell sampling. When areas impacted by either the old homestead or animals were sampled separately, splitting the field into 4-ha grid cells reduced underfertilized areas from 63.3 to 34.8% and increased overfertilized areas from 30.0 to 32.6% (Table 9). These results show that the net effect of grid-cell sampling at Moody was to reduce underfertilized areas with a minimal impact on overfertilized areas. The amount of land over- and underfertilized was similar when the old homestead or animal-impacted areas were included or not included in the data set.

At Brookings, removing the old homestead from the data set had a relatively small impact on total N applied (<3%). Grid-cell sampling (3.5 ha) reduced the percentage of land over- and underfertilized. At Beresford, when areas impacted by an old homestead was removed from the data set, grid-cell sampling (4 ha) reduced underfertilized areas from 51.5 to 36.4%, and did not influence the total N recommendation or the percentage of the areas overfertilized.

Grid-cell sampling is not new and has been compared with several other approaches for obtaining spatial in-

Table 5. The influence of sampling the homestead or grazing area separately from the rest of the field and separating the field into 3.5- to 4-ha blocks on the percentage of the three fields that would have been over- and underfertilized with P fertilizer.

Sampling methods	Old homestead or animal-impacted areas					
	Sampled separately			Not sampled separately		
	Areas of fertilized		P recommended	Areas of fertilized		P recommended
	Over	Under		Over	Under	
	———— % ————			———— % ————		
		kg P per field			kg P per field	
Moody						
4-ha grid cell	27.6	45.0	764	23.1	46.1	708
Whole field	31.5	57.1	720	31.7	63.6	516
Brookings						
3.5-ha grid cell	25.1	37.6	228	25.1	37.8	220
Whole field	0	55.2	0	0	55.2	0
Beresford						
4-ha grid cell	16	39	749	14	40	629
Whole field	30	63	401	0	72	0

Table 6. The influence of sampling approach on $\text{NO}_3\text{-N}$ (mg kg^{-1}) pooled variances (s_p^2) at Moody. The information used in classification approaches were combinations of apparent electrical conductivity (EC_a), elevation (elev.), distance, and aspect.

Sampling methods	Classes (<i>n</i>)	Old homestead or animal-impacted area			
		Sampled separately		Not sampled separately	
		s_p^2	<i>F</i> value	s_p^2	<i>F</i> value
Grid cell					
4-ha grid cell	16	38.1	1.96†	36.3	2.00†
7-ha grid cell	9	48.3	1.54†	46.0	1.58†
16-ha grid cell	4	44.0	1.70†	41.9	1.73†
GIS					
EC_a -elev.	10	66.3	1.12	64.1	1.13
EC_a -aspect	18	67.8	1.10	65.4	1.11
EC_a -distance	15	57.1	1.31†	54.3	1.34†
Cluster					
EC_a -elev.	4	68.3	1.09	66.5	1.09
EC_a -aspect	2	71.6	1.04	69.1	1.05
EC_a -elev.-aspect	5	71.6	1.04	69.8	1.04
Order 1 soil survey	16	72.1	1.03	70.1	1.04
Whole field	1	74.6		72.6	

† s_p^2 values are significantly different from the whole field at ($P = 0.1$). For the numerator, $df = 536$. For the denominator, $df = 536 - n$.

formation. Mueller et al. (2001) reported that kriged maps based on a 100-m grid, grid cell, and simulated soil map units has similar prediction efficiencies, which were poor. Wollenhaupt et al. (1994) compared grid cell vs. grid-point sampling for P and K recommendations in two Wisconsin fields. Wollenhaupt et al. (1994) recommended that if (i) a single rate of fertilizer was to be applied, then grid-cell sampling is better than whole-field sampling; (ii) the soil test P and K levels are in the nonresponsive categories, then the field should be sampled on a 91-m grid; and (iii) the field is in the response category, then the field should be sampled on a 61-m grid. Wollenhaupt et al. (1994) and this study had one major difference: the interpretation. In Wisconsin, yields are higher than those in the northern Great Plains, and therefore the potential to recover sampling costs are greater in Wisconsin than the northern Great Plains. On the basis of the recommendations of Wollenhaupt et al. (1994), approximately 64 and 144 samples would be collected from fields sampled on a 91- and 61-m grids. Many farmers in the northern Great Plains are not willing to collect this number of samples because they perceive this as unprofitable.

SUMMARY

This study investigated the impact of different techniques for identifying N and P management zones on s_p^2 . When compared with the whole field, soil sampling based on the Order 1 soil survey did not reduce the Olsen-P and $\text{NO}_3\text{-N}$ s_p^2 . The failure of the Order 1 soil survey to reduce Olsen-P and $\text{NO}_3\text{-N}$ s_p^2 was attributed to the impact of prior management on nutrient variability.

Generally, the lowest s_p^2 values resulted from grid-cell sampling or areas impacted by an old homestead and animals separately from the rest of the field. The net effect of 4-ha (3.5-ha) grid-cell sampling when compared with whole-field sampling was to reduce the percentage of the land underfertilized. These results were attributed to Olsen-P and $\text{NO}_3\text{-N}$ having strong to moderate spatial dependence (Table 1). Spatial structure was important because in grid-cell sampling, distance is the only criterion used for identifying the zones. Sampling zones based on grid-cell sampling (3.5 and 4 ha) had shorter distances between all potential sampling points than any other technique tested. For example, in the 4-ha grid cells, the maximum distance between potential sampling

Table 7. The influence of sampling approach on $\text{NO}_3\text{-N}$ (mg kg^{-1}) pooled variances (s_p^2) at Brookings, SD. The information used in classification approaches were combinations of apparent electrical conductivity (EC_a), elevation (elev.), distance, and aspect.

Sampling methods	Classes (<i>n</i>)	Old homestead or animal-impacted areas			
		Sampled separately		Not sampled separately	
		s_p^2	<i>F</i> value	s_p^2	<i>F</i> value
Grid cell					
3.5-ha grid cell	14	5.38	1.18†	6.85	1.18†
6.6-ha grid cell	8	5.61	1.13	7.08	1.03
GIS					
EC_a -elev.	9	5.84	1.09	7.62	1.06
EC_a -aspect	13	6.36	1.00	7.64	1.06
EC_a -distance	10	6.23	1.02	7.85	1.03
Cluster					
EC_a -elev.	8	6.14	1.03	7.78	1.04
EC_a -aspect	9	6.16	1.03	7.75	1.04
EC_a -elev.-aspect	8	6.26	1.01	7.93	1.02
Order 1 soil survey	8	6.33	1.00	7.96	1.02
Whole field	1	6.34		8.10	

† s_p^2 values are significantly different from whole field at ($P = 0.1$). For the numerator, $df = 352$. For the denominator, $df = 352 - n$.

Table 8. The influence of sampling approach on $\text{NO}_3\text{-N}$ (mg kg^{-1}) pooled variances (s_p^2) at Beresford, SD. The information used in classification approaches were combinations of apparent electrical conductivity (EC_a), elevation (elev.), distance, and aspect.

Sampling methods	Classes (n)	Old homestead or animal-impacted areas			
		Sampled separately		Not sampled separately	
		s_p^2	F value	s_p^2	F value
Grid cell					
4-ha grid cell	10	13.9	1.31†	47.3	1.14
7-ha grid cell	6	15.9	1.14	51.0	1.06
13-ha grid cell	3	17.6	1.03	54.9	0.99
GIS					
EC_a -elev.	8	15.2	1.19	52.9	1.02
EC_a -aspect	11	16.5	1.10	29.7	1.82†
EC_a -distance	11	16.1	1.12	42.7	1.27
Cluster					
EC_a -elev.	8	15.6	1.16	45.7	1.18
EC_a -aspect	5	17.4	1.04	45.2	1.20
EC_a -elev.-aspect	9	15.7	1.16	44.9	1.20
Whole field	1	18.1		54.0	

† s_p^2 values are significantly different from whole field at ($P = 0.1$). For the numerator, $df = 100$. For the denominator, $df = 100 - n$.

Table 9. The influence of sampling the homestead or grazing area separately from the rest of the field and separating the field into 3.5- to 4-ha blocks on the percentage of the three fields that would have been over- and underfertilized with N fertilizer.

Sampling methods	Old homestead or animal-impacted areas					
	Sampled separately			Not sampled separately		
	Areas of fertilized		N recommended	Areas of fertilized		N recommended
	Over	Under		Over	Under	
	%		kg per field	%		kg per field
Moody						
4-ha grid cell	32.6	34.8	2783	33.5	35.6	3070
Whole field	30.0	63.3	1980	30.5	60.9	2154
Brookings						
3.5-ha grid cell	18.8	27.8	5584	19.9	30.7	5757
Whole field	20.5	30.7	5576	20.2	33.8	5514
Beresford						
4-ha grid cell	32.3	36.4	2429	33.3	48.5	2446
Whole field	32.3	51.5	2428	33.3	66.7	2120

points was 283 m, while distances between potential sampling points for the Order 1 soil survey could be >800 m. Although not always significant, management zones based on EC_a -distance information for soil test P (two of three sites) and nitrate-N (one of three sites) had lower s_p^2 than classification approaches that did not consider distance. These results were attributed to EC_a -distance classification separating areas with similar EC_a values that were not physically connected to each other into two different zones. These results point out the importance of distance between sampling points, and suggest that soil samples collected at similar landscape positions that are separated by long distances should not be combined.

The effect of sampling areas impacted by homesteads or animals separately from the rest of the field indicates that events of 30 to 50 yr ago still affect nutrient variability today. Many current soil sampling guidelines suggest that these areas should be sampled separately from the rest of the field (Jacobsen, 1998; Buchholz, 1999; Kleinjan, 2002). Results from this study confirm these recommendations. Findings from this study suggested that (i) soil sampling variation can be reduced by grid-cell sampling; (ii) old homestead or grazing areas should be sampled separately from the rest of the field; and (iii) zones that are not continuous should be sampled and managed separately.

ACKNOWLEDGMENTS

Support for this project came in part from the South Dakota Soybean Research and Promotion Council, United Soybean Board, North Central Soybean Research Program, USDA-CSREES, and NASA.

REFERENCES

- Buchholz, D.D. 1999. How to get a good soil sample [Online]. Missouri Ext. Publ. G09110. Available at: <http://muextension.missouri.edu/xplor/agguides/soils/g09110.htm> [posted 1 Oct. 1993; cited 6 May 2003; verified 21 Aug. 2003]. Dep. of Agronomy, Univ. of Missouri, Columbia.
- Cambardella, C.A., T.B. Moorman, J.M. Novak, T.B. Parkin, D.L. Karlan, R.F. Turco, and A.E. Konopka. 1994. Field-scale variability of soil properties in central Iowa soils. *Soil Sci. Soc. Am. J.* 58: 1501–1511.
- Chang, J., D.E. Clay, C.G. Carlson, S.A. Clay, and C.L. Reese. 2000. The influence of different approaches for identifying inorganic N and P management zones on fertilizer recommendation [CD-ROM computer file]. In P.C. Robert et al. (ed.) *Proc. of the 5th Int. Conf. on Precision Agriculture*, Bloomington, MN. 16–19 July 2000. ASA, CSSA, and SSSA, Madison, WI.
- Chang, J., D.E. Clay, C.G. Carlson, J. Lee, D.D. Malo, S.A. Clay, and M. Ellsbury. 1999. Selecting precision farming sampling protocols: Part 1. Grid distance impact on semivariogram and estimation variances. *Precis. Agric.* 1:277–289.
- Clay, D.E., C.G. Carlson, K. Brix-Davis, J. Oolman, and B. Berg. 1997. Soil sampling strategies for estimating residual nitrogen. *J. Prod. Agric.* 10:446–452.
- Clay, D.E., J. Chang, D.D. Malo, C.G. Carlson, C.L. Reese, S.A.

- Clay, M. Ellsbury, and B. Berg. 2001. Factors influencing spatial variability of soil apparent electrical conductivity. *Commun. Soil Sci. Plant Anal.* 32:2993–3008.
- Doerge, T.A. 1999. Management zone concepts. SSMG-2. In D.E. Clay et al. (ed.) *Site specific management guidelines* [Online]. Available at: <http://ppi-far.org/ssmg> [cited 17 Sept. 2002, verified 21 Aug. 2003]. Potash and Phosphate Inst., Norcross, GA.
- Environmental Systems Research Institute. 1996. Using ArcView. ESRI, Redland, CA.
- Ferguson, R.B., K.D. Frank, G.W. Hergert, E.J. Pena, and R.A. Wiese. 1998. Guidelines for soil sampling [Online]. Publ. G91-1000-A. Available at: <http://www.ianr.unl.edu/pubs/Soil/g1000.htm> [posted February 1991; cited 6 Jan. 2003; verified 21 Aug. 2003]. Cooperative Ext. Serv., Inst. of Agric. and Nat. Resources, Univ. of Nebraska, Lincoln.
- Fleming, K.L., D.G. Westfall, and W.C. Bausch. 2000. Evaluating management zone technology and grid soil sampling for variable rate nitrogen application [CD-ROM computer file]. In P.C. Robert et al. (ed.) *Proc. of the 5th Int. Conf. on Precision Agriculture*, Bloomington, MN. 16–19 July 2000. ASA, CSSA, and SSSA, Madison, WI.
- Fleming, K.L., D.G. Westfall, D.W. Wiens, L.E. Rothe, J.E. Cipra, and D.F. Heermann. 1999. Evaluating farmer developed management zone maps for precision farming. p. 335–343. In P.C. Robert et al. (ed.) *Proc. of the 4th Int. Conf. on Precision Agriculture*, St. Paul, MN. 19–22 July 1998. ASA, CSSA, and SSSA, Madison, WI.
- Franzen, D.W., L.J. Cihacek, V.L. Hofman, and L.J. Swenson. 1998. Topography-based sampling compared with grid sampling in the Northern Great Plains. *J. Prod. Agric.* 11:364–370.
- Franzen, D.W., D.H. Hopkins, M.D. Sweeney, M.K. Ulmer, and A.D. Halvorson. 2002. Evaluation of soil survey scale for zone development of site-specific nitrogen management. *Agron. J.* 94:381–389.
- Fridgen, J.J. 2000. Development and evaluation of unsupervised clustering software for sub-field delineation of agricultural fields. M.S. thesis. Univ. of Missouri, Columbia.
- Fridgen, J.J., N.R. Kitchen, and K.A. Sudduth. 2000. Variability of soil and landscape attributes within sub-field management zones. [CD-ROM computer file]. In P.C. Robert et al. (ed.) *Proc. of the 5th Int. Conf. on Precision Agriculture*. 16–19 July 2000. Bloomington, MN. ASA, CSSA, and SSSA, Madison, WI.
- Gerwing, J., and R. Gelderman. 1998. Fertilizer recommendations guide (EC 750) [Online]. Available at: <http://plantsci.sdstate.edu/woodardh/soiltest/> [posted March 1998; cited May 2002; verified 21 Aug. 2003]. Cooperative Ext. Serv., South Dakota State Univ., Brookings.
- Jacobsen, J.J. 1998. Soil sampling [Online]. Publ. no. 8602 Agriculture. Available at: <http://www.montana.edu/wwwpb/pubs/mt8602.html> [verified 21 Aug. 2003]. MSU Ext. Serv., Montana State Univ., Bozeman, MT.
- Johnson, D. 1998. Applied multivariate methods for data analysis. Duxbury Press, Pacific Grove, CA.
- Johansen, D.P., D.E. Clay, C.G. Carlson, K.W. Stange, S.A. Clay, D.D. Malo, and J.A. Schumacher. 2001. Vertical accuracy of two differentially corrected global positioning systems. *J. Soil Water Conserv.* 56:198–201.
- Kleinjan, J. 2002. Previous management impacts on soil phosphorus levels. M.S. Thesis. South Dakota State Univ., Brookings.
- Kravchenko, A., and D.G. Bullock. 1999. A comparative study of interpolation methods for mapping soil properties. *Agron. J.* 91:393–400.
- Mallarino, A.P., and D.J. Wittry. 2001. Identifying cost-effective soil sampling schemes for variable-rate fertilization and liming [CD-ROM computer file]. In P.C. Robert et al. (ed.) *Proc. of the 5th Int. Conf. on Precision Agriculture*, Bloomington, MN. 16–19 July 2000. ASA, CSSA, and SSSA, Madison, WI.
- Maynard, D.G., and Y.P. Kalra. 1993. Nitrate and exchangeable ammonium nitrogen. p. 25–38. In M.R. Carter (ed.) *Soil sampling and methods of analysis*. Lewis Publ., Boca Raton, FL.
- Mueller, T.G., F.J. Pierce, O. Schabenberger, and D.D. Warncke. 2001. Map quality for site-specific fertility management. *Soil Sci. Soc. Am. J.* 65:1547–1558.
- Murphy, J., and H.P. Riley. 1962. A modified single solution method for the determination of phosphate in natural waters. *Anal. Chim. Acta* 27:31–36.
- Olsen, S.R., and L.E. Sommers. 1982. Phosphorus. p. 403–430. In A.L. Page (ed.) *Methods of soil analysis*. Agron. Monogr. 9. Part 2, 2nd ed. ASA and SSSA, Madison, WI.
- Ott, L. 1977. An introduction to statistical methods and data analysis. Duxbury Press, Belmont, CA.
- Soil Survey Staff. 1993. Soil survey manual. USDA Handbk. 18. U.S. Gov. Print. Office, Washington, DC.
- Steel, R.G.D., and J.H. Torrie. 1980. Principles and procedures of statistics. 2nd ed. McGraw-Hill, New York.
- USDA-ARS. 2000. Management zone analyst 1.0 [Online]. Available at: <http://www.fse.missouri.edu/mpac/whatsnew.htm> (updated 8 Feb. 2002; cited 6 Jan. 2003; verified 21 Aug. 2003). Missouri Precision Agriculture Center, University of Missouri, Columbia.
- Wollenhaupt, N.C., R.P. Wolkowski, and M.K. Clayton. 1994. Mapping soil test phosphorus and potassium for variable-rate fertilizer application. *J. Prod. Agric.* 7:441–448.